GIT Bleeding

prepared by

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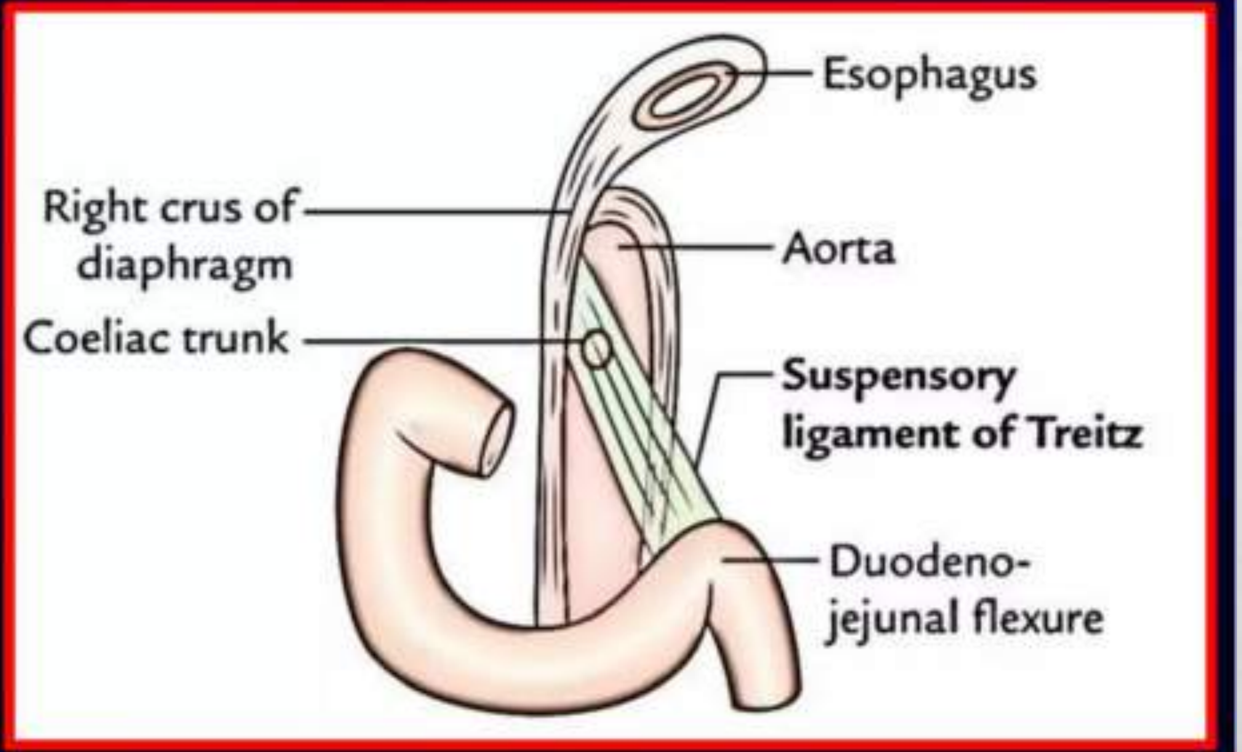
Upper Gastrointestinal bleeding

- Is hemorrhage originating above the ligament of Treitz, which lies at the duodenojejunal flexure
- It is the most common git emergency
- It is 4 times more common than lower git bleeding
- The mortality rate in hospitalized patients is about 5-10%



Causes of upper gastrointestinal bleeding

- **Common**

1. Peptic ulceration.
2. Erosion
3. Mucosal inflammation (esophagitis, gastritis, or duodenitis).
4. esophageal and gastric varices.
5. Mallory–Weiss tear.
6. Tumors
7. Bleeding tendency

- **Rare**

1. Aorto-enteric fistula
2. Vascular lesions, e.g. Dieulafoy's disease

❖ It is important to classify upper git bleeding into variceal and non variceal bleeding because the management is different.

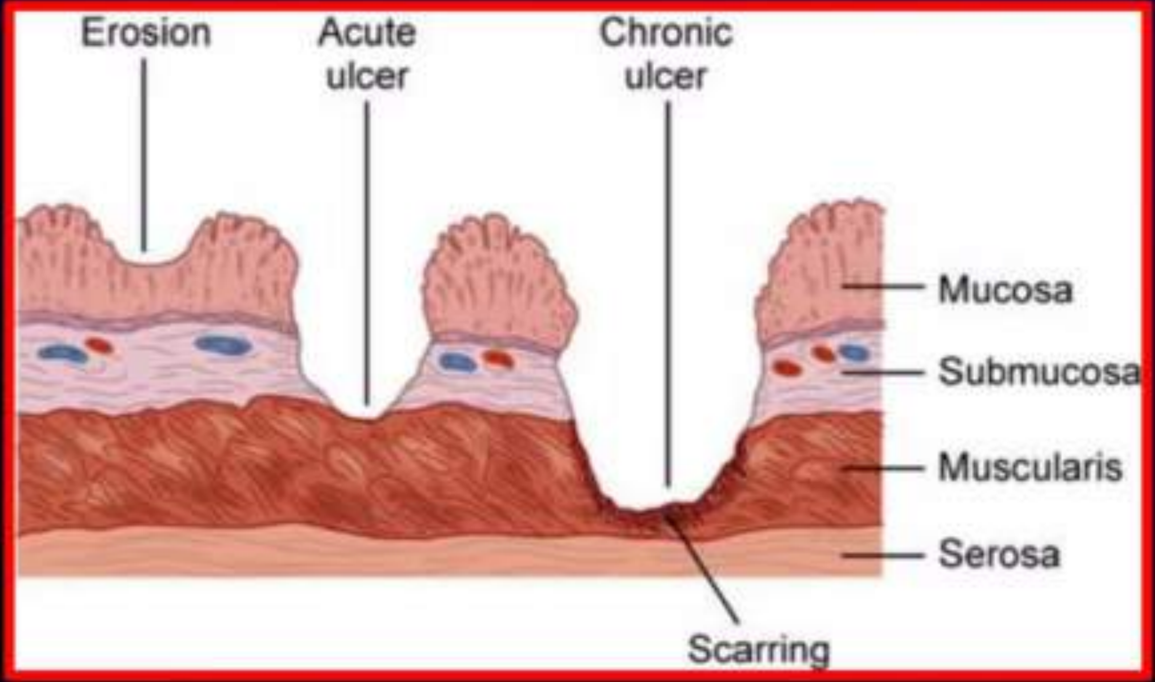
Peptic ulcer

- peptic ulcer refers to an ulcer in the lower esophagus, stomach, duodenum or in the jejunum after surgical anastomosis to the stomach
- Peptic ulceration affects areas of mucosa exposed to acidic gastric contents. The main pathology is an imbalance between the acid–pepsin system and the mucosal ability to resist digestion.
- Duodenal ulcers occur four times more commonly than gastric ulcers and are more likely to bleed.

Aetiology

- Helicobacter pylori
- Nonsteroidal anti-inflammatory drugs
- Smoking

Erosions are similar to peptic ulcer but do not penetrate the muscularis Mucosae and are usually caused by NSAIDs and alcohol



Mucosal inflammation

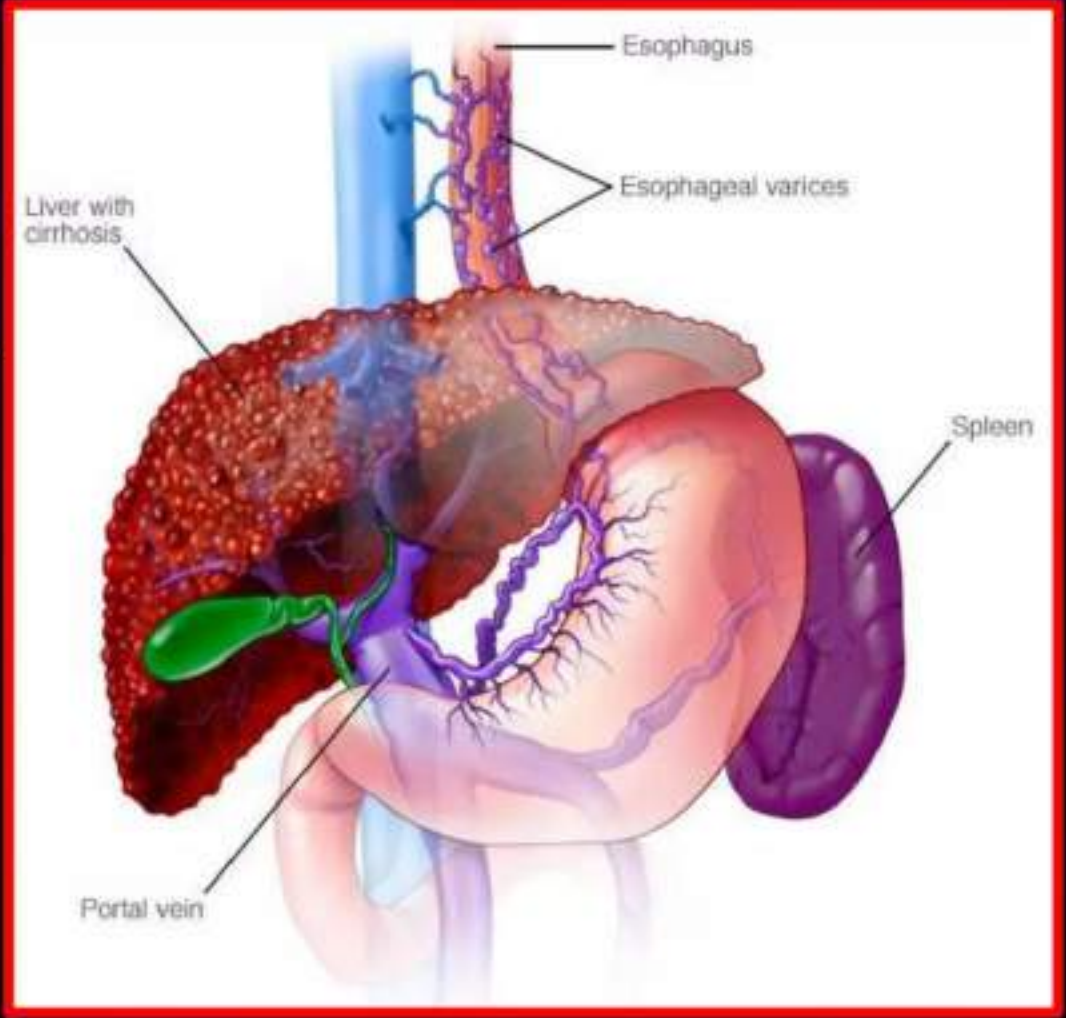
1. **Esophagitis** : reflux, infective, chemical, eosinophilic
2. **Gastritis:**

- **Acute:** Aspirin, NSAIDs, Severe physiological stress, e.g. burns, multi-organ failure, Alcohol, Bile reflux, Viral infections.
- **Chronic:** H. pylori infection, Autoimmune (pernicious anemia), Infections, Gastrointestinal diseases (Crohn's disease), Systemic diseases (sarcoidosis, graft-versus-host disease).

3. **Duodenitis**

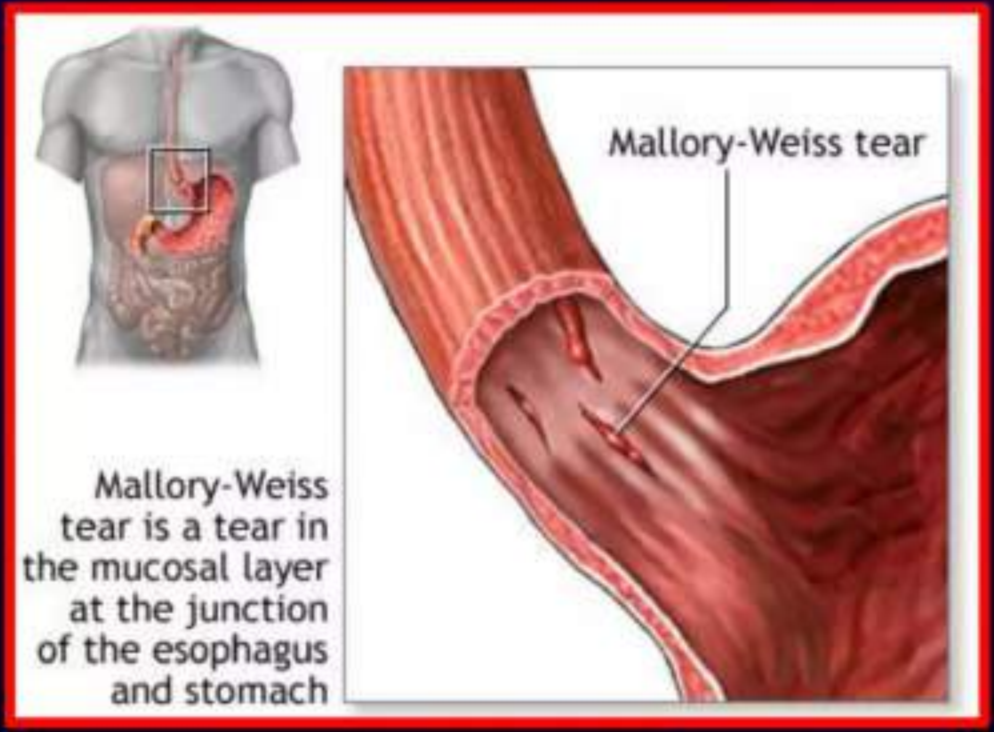
Portal hypertension and bleeding varices

- Increased portal vascular resistance leads to the development of collateral vessels, allowing portal blood to bypass the liver and enter the systemic circulation directly. Portosystemic shunting occurs, particularly in the gastrointestinal tract and especially the distal esophagus and stomach.
- Esophageal varices are located within 3–5 cm of the gastro-esophageal junction.
- Variceal bleeding is often severe, and recurrent if preventative treatment is not given.



MALLORY-WEISS TEAR

- a longitudinal tear at the to the esophageal mucosa at the gastro-esophageal junction, which is induced by repetitive and strenuous vomiting and retching.
- Most patients present with hematemesis. Bleeding usually is not severe and stops spontaneously.
- protracted bleeding may respond to local epinephrine or cauterization therapy, endoscopic clipping, or angiographic embolization.
- Surgery is rarely needed.



Other causes

- **Cancer** : Adenocarcinoma is the most common malignancy, bleeding is a frequent symptom especially in advanced tumors.
- **Bleeding tendency** : (anticoagulant drugs , bleeding disorders)
- **Aorto-enteric fistula** (especially after aortic surgery).
- **Dieulafoy's disease**: gastric arterial venous malformation that has a characteristic histological appearance.

AN APPROACH TO PT. WITH UPPER GI BLEEDING

- Initial Assessment and Resuscitation
- History and Physical Examination
- Assessment of the bleeding source
- Differential Diagnosis
- Investigations
- Management

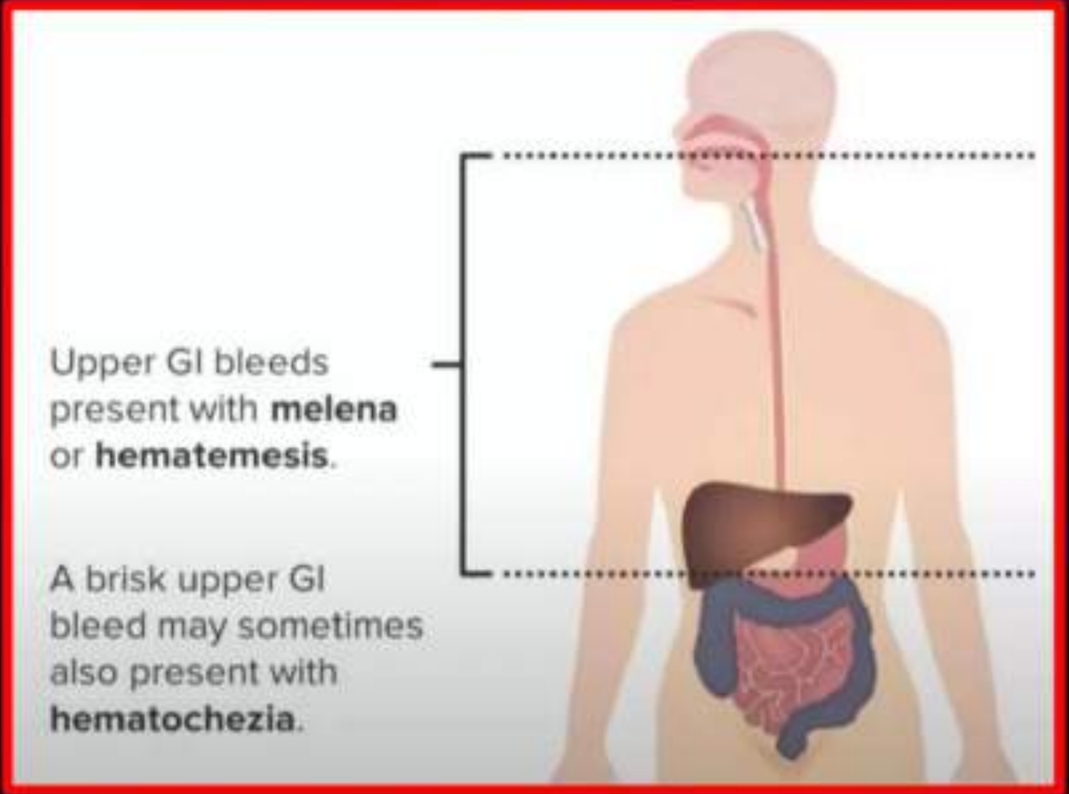
Initial Assessment and Resuscitation

- **Airway, Breathing and Circulation**
- **Vital Signs:**
- Pulse, BP, Temperature, Respiratory Rate
- **Fluid and Resuscitation Plan**
- The first step is to gain intravenous access using two wide-bore cannula.
- Intravenous crystalloid fluids should be given to raise the blood pressure, and blood should be transfused when the patient is actively bleeding with low blood pressure and tachycardia.
- **Oxygen**
- This should be given to all patients in shock.

History and Physical Examination

• Manner of Presentation

1. Hematemesis
2. Malena
3. Hematochezia
4. Occult Blood loss



History and Physical Examination

- **Symptoms of Blood loss: include those suggestive of acute substantial blood loss and shock (hypotension, tachycardia, tachypnoea and pallor)**
- **Symptoms that suggest the bleeding is severe: include orthostatic dizziness, confusion, angina, severe palpitations, and cold/clammy extremities.**

History and Physical Examination

Past History

- Similar episodes before
- H/o Jaundice in past
- H/o Abdominal Surgery

social history

- H/o Alcoholism
- H/o Smoking or Tobacco abuse

History and Physical Examination

Medication history

- Predispose to peptic ulcer formation, such as aspirin and other NSAIDs, including COX-2 inhibitors
- Anticoagulation (warfarin/heparin)
- Promote bleeding, such as antiplatelet agents (e.g. clopidogrel) and anticoagulants (including the direct oral anticoagulants)
- Have been associated with GI bleeding, including selective serotonin reuptake inhibitors (SSRI), calcium channel blockers, and aldosterone antagonists
- iron supplement , which can turn the stool black

Physical Examination

- Pt's Consciousness, Orientation
- Pallor, Clubbing, Peripheral Edema
- Signs of Liver Failure
- Systemic Examination : Abdomen, CVS, RS, CNS

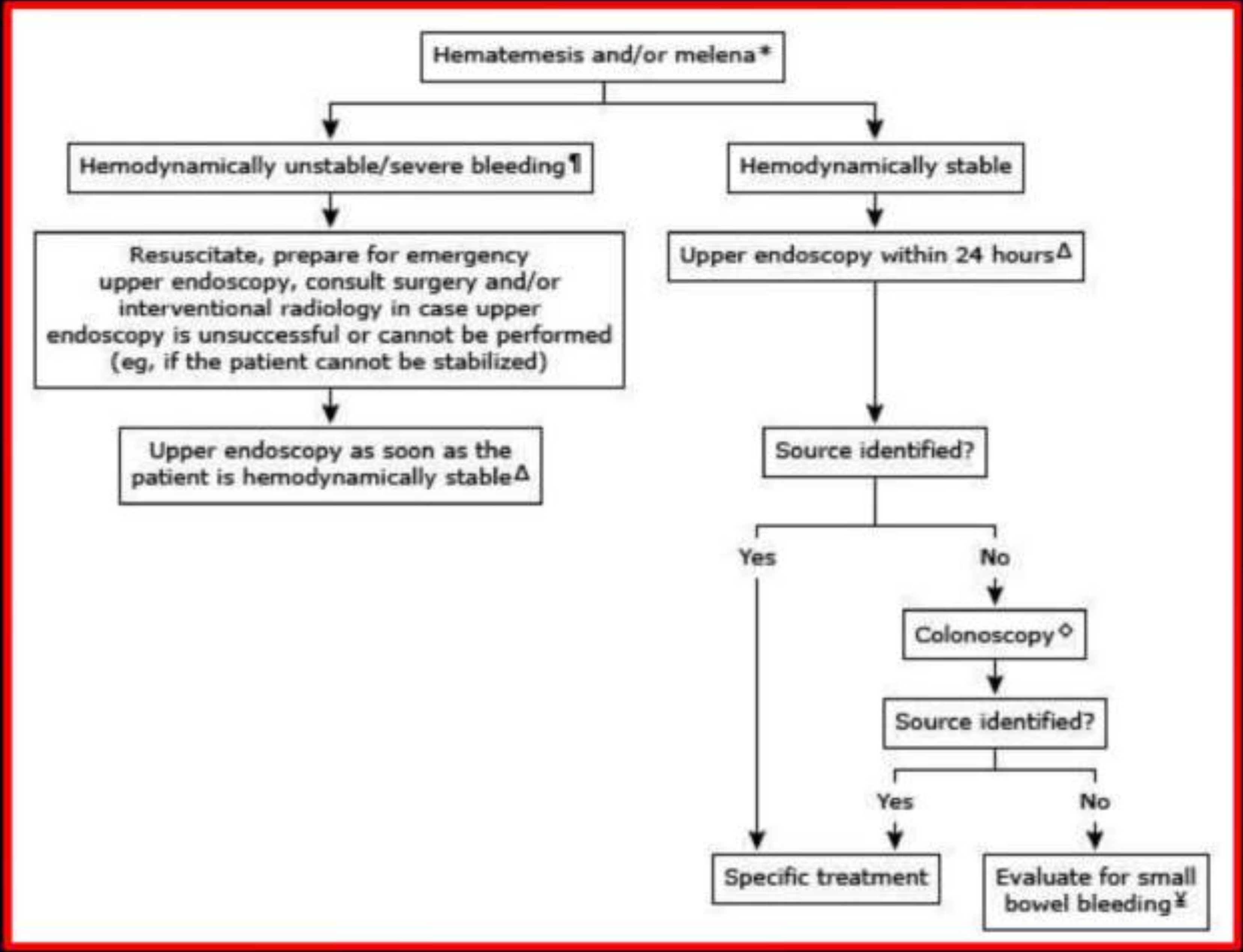
- **Specific features and signs of liver disease and portal hypertension (spider naevi, portosystemic shunting and bruising).**

Specific causes of upper GI bleeding may be suggested by the patient's symptoms:

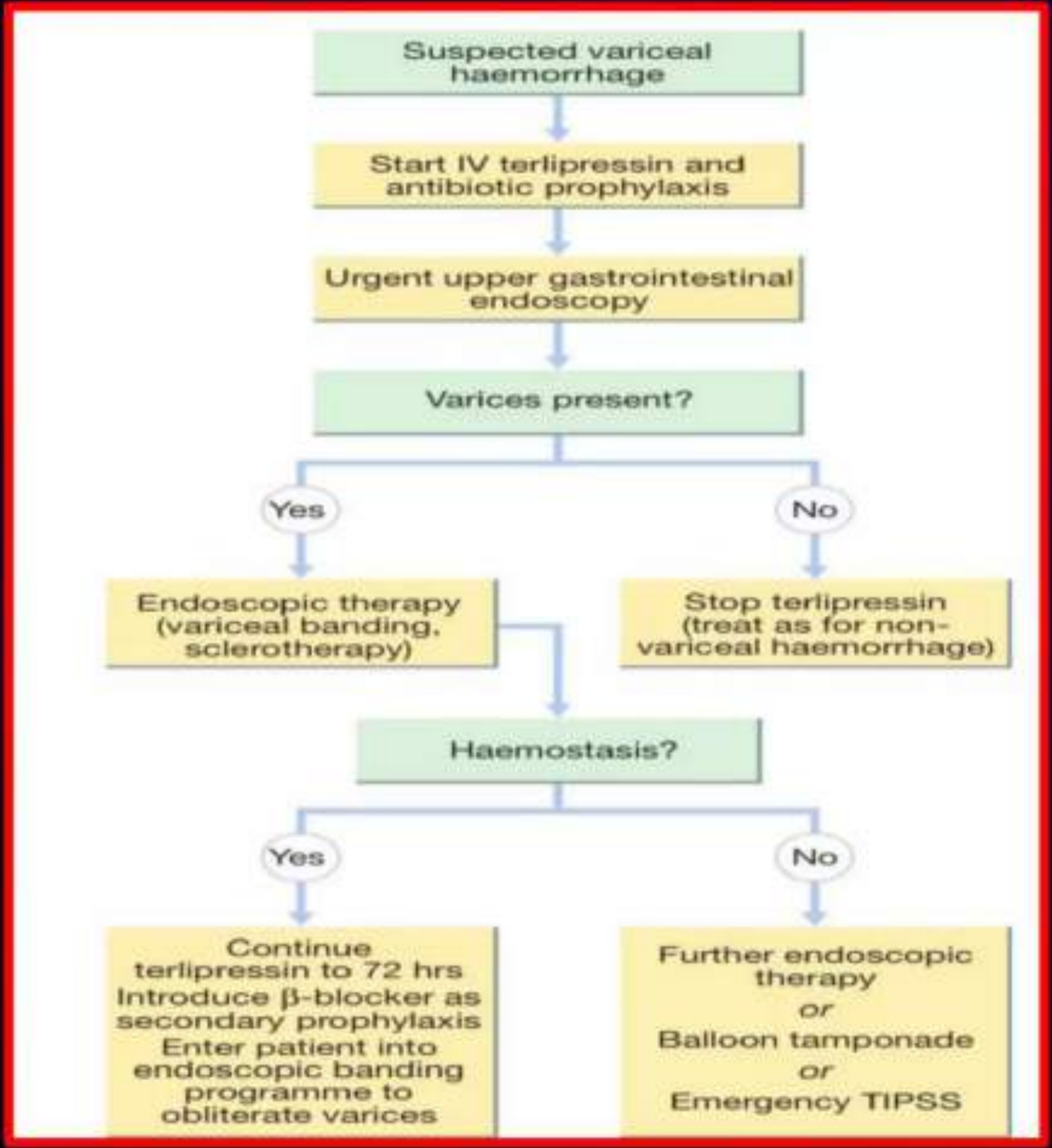
- **Peptic ulcer** – Upper abdominal pain
- **Esophageal ulcer** – Odynophagia, gastroesophageal reflux, dysphagia
- **Mallory-Weiss tear** – Emesis, retching, or coughing prior to hematemesis
- **Variceal hemorrhage or portal hypertensive gastropathy**: Jaundice, abdominal distention (ascites)
- **Malignancy** – Dysphagia, early satiety, involuntary weight loss, cachexia

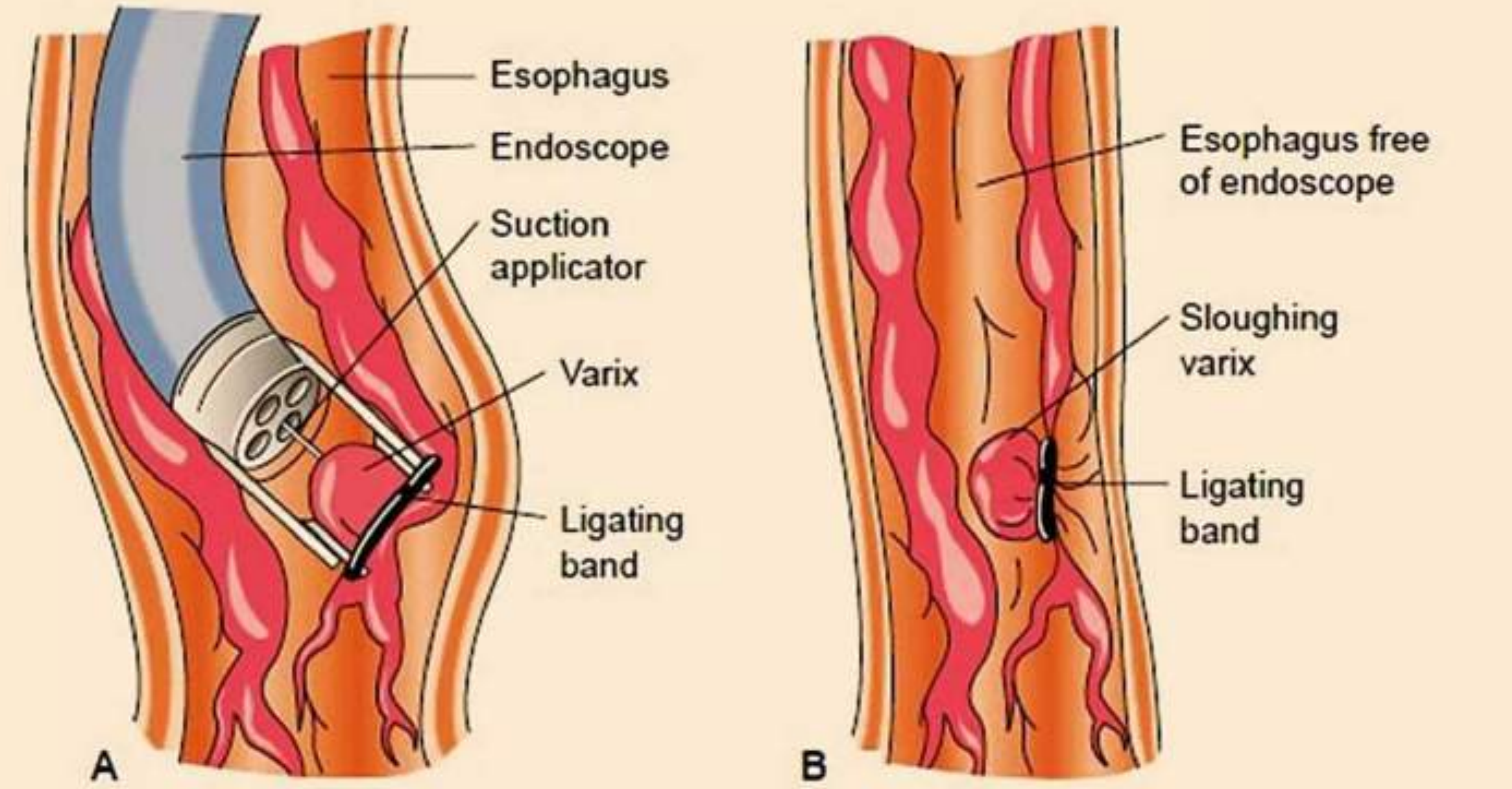
Basic investigations

- **Full blood count.** Chronic or subacute bleeding leads to anemia, but the hemoglobin concentration may be normal after sudden, major bleeding until hemodilution occurs. Thrombocytopenia may be a clue to the presence of hypersplenism in chronic liver disease.
- **Urea and electrolytes.** This test may show evidence of renal failure. The blood urea rises as the absorbed products of luminal blood are metabolised by the liver; an elevated blood urea with normal creatinine concentration implies severe bleeding.
- **Liver function tests.** These may show evidence of chronic liver disease.
- **Prothrombin time.** Check with clinical suggestion of liver disease or in anticoagulated patients.
- **Crossmatching.** At least 2 units of blood should be cross-matched.


Hematemesis and/or melena*
Hemodynamically unstable/severe bleeding¶
Hemodynamically stable
Resuscitate, prepare for emergency upper endoscopy, consult surgery and/or interventional radiology in case upper endoscopy is unsuccessful or cannot be performed (eg, if the patient cannot be stabilized)
Upper endoscopy within 24 hoursΔ
Upper endoscopy as soon as the patient is hemodynamically stableΔ
Source identified?
Yes
No
Colonoscopy◊
Source identified?
Yes
No
Specific treatment
Evaluate for small bowel bleeding¥

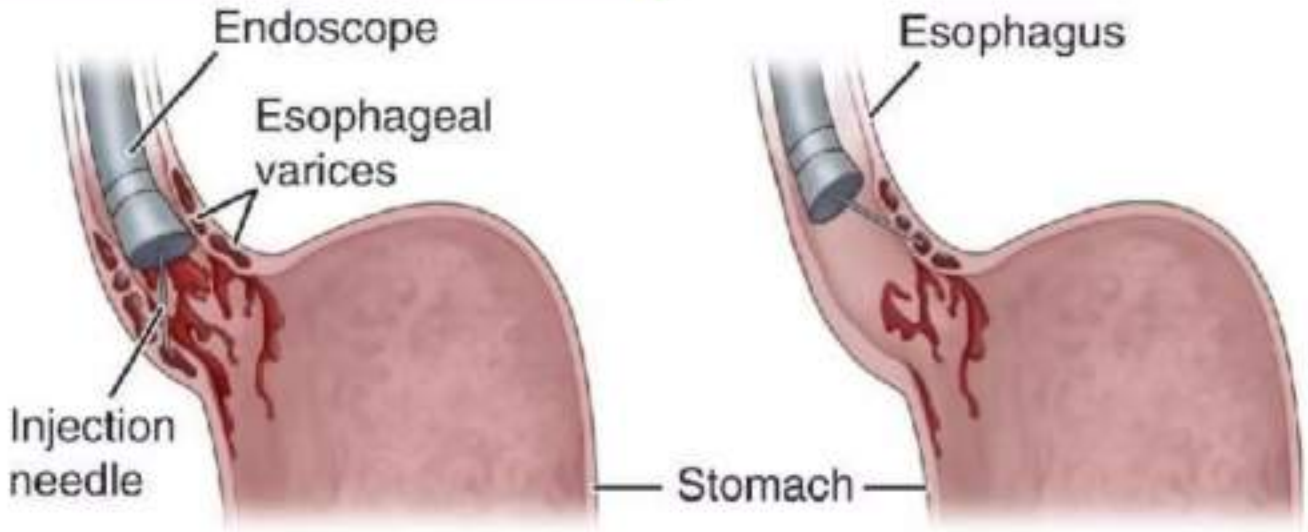
Management of variceal bleeding



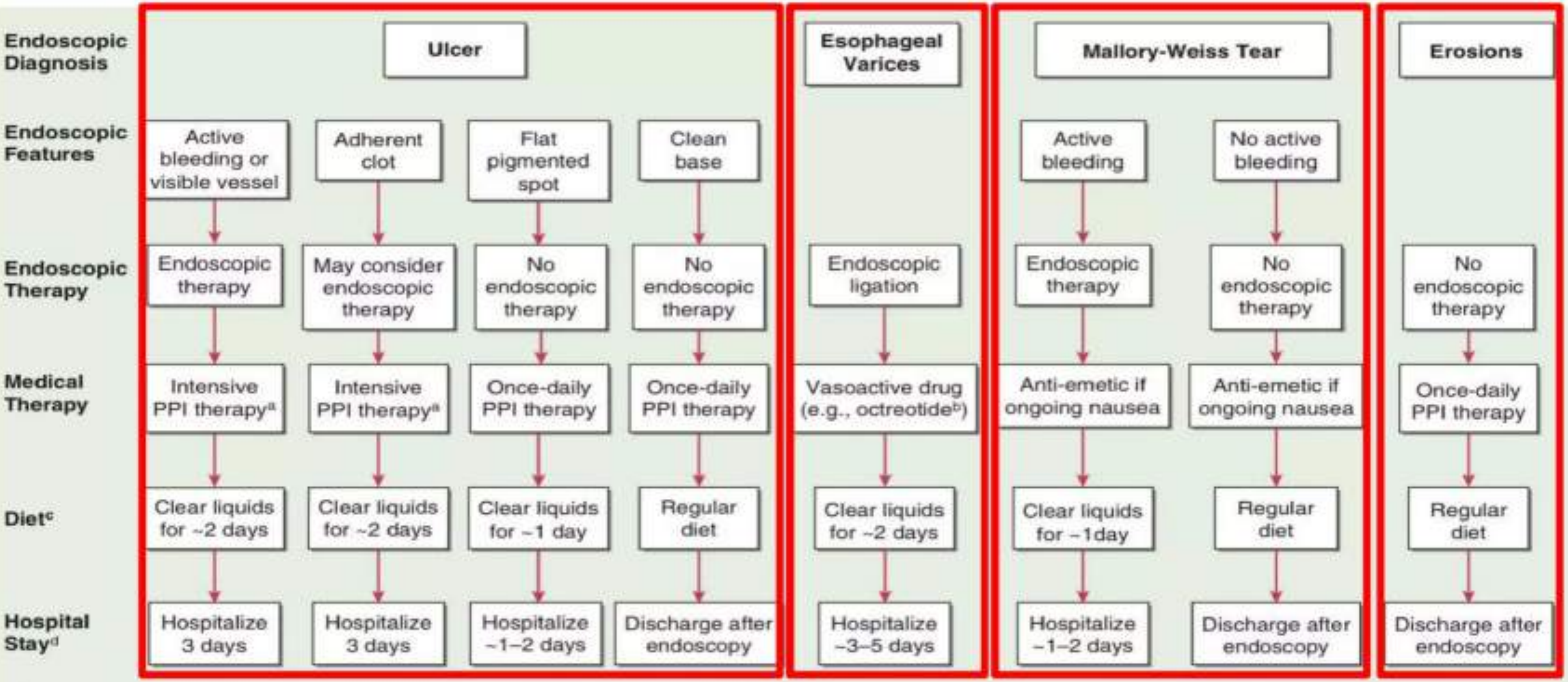


ENDOSCOPIC VARICEAL BAND LIGATION

Endoscopic Sclerotherapy



Management of non-variceal hemorrhage



[a]Intravenous bolus (80 mg) followed by infusion (8 mg/h) for 3 days; or oral or intravenous bolus (e.g., 80 mg) followed by intermittent high doses (e.g., 40–80 mg bid or 40 mg tid) for 3 days. Then twice-daily PPI on days 4–14 followed by once-daily PPI.

[b]Intravenous 50 μg bolus followed by 50 μg/h infusion for 2–5 days.

[c]Diet after endoscopy, assuming no nausea or vomiting.

[d]Duration after endoscopy assuming patient stable without further bleeding or concurrent medical conditions requiring hospitalization; PPI, proton pump inhibitor.

FIGURE 44-1 Suggested algorithm for patients with acute upper gastrointestinal bleeding based on endoscopic findings.

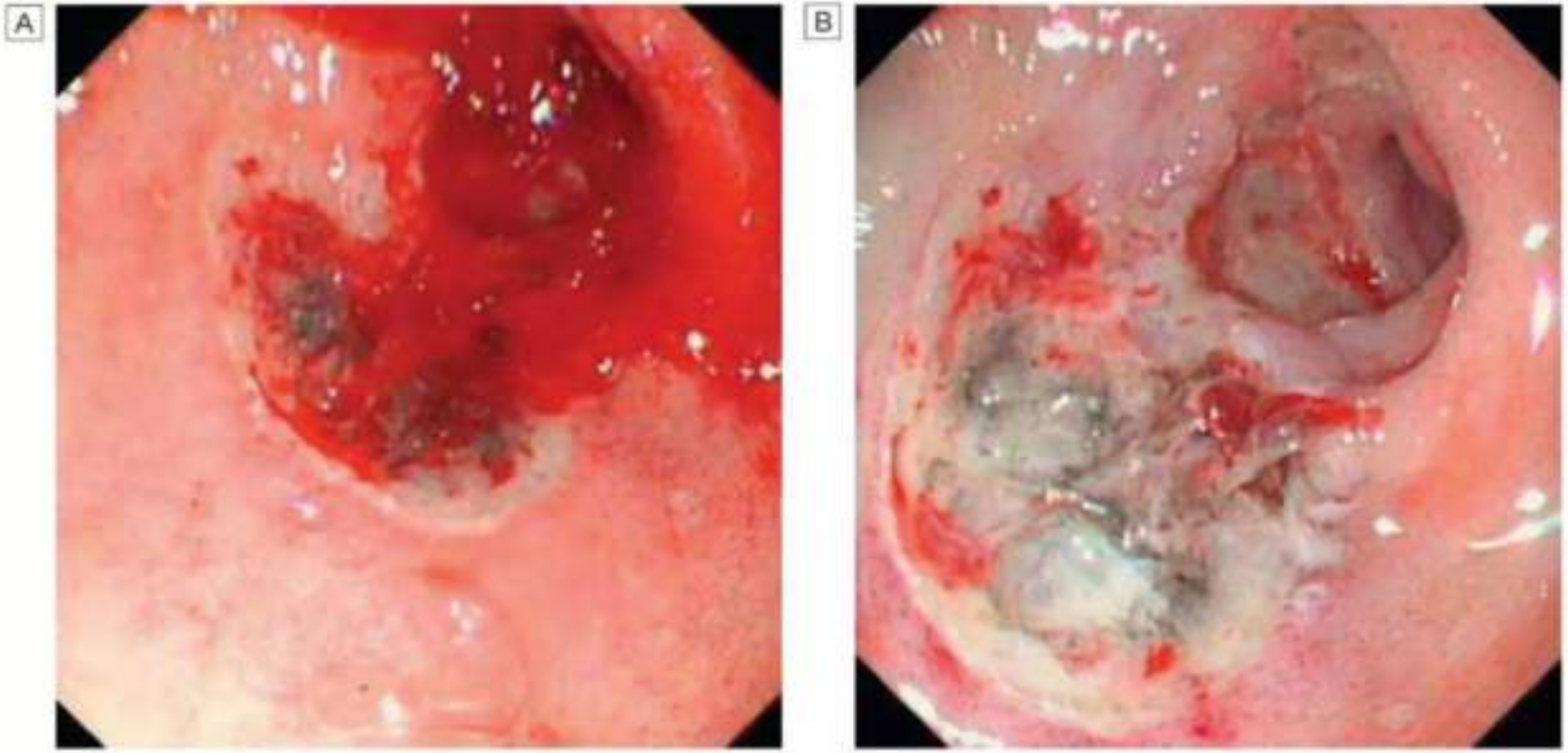


Fig. 21.20 Major stigmata of recent haemorrhage and endoscopic treatment. A Active bleeding from a duodenal ulcer. B Haemostasis is achieved after endoscopic injection of adrenaline (epinephrine) and application of a heater probe.

Indication of surgery

1. Failure of medical treatment and endoscopic homeostasis
2. Persistent hypotension
3. Coexisting condition (perforation, obstruction, malignancy)

TYPES OF OPERATIONS

The choice of operation depends on the site and the bleeding lesions:

1. Duodenal ulcers are treated by under-running with or without pyloro-plasty.
2. Gastric ulcers treated by under-running (take a biopsy to exclude carcinoma).
3. Local excision or partial gastrectomy may be required.


THANK
YOU

Lower GIT Bleeding

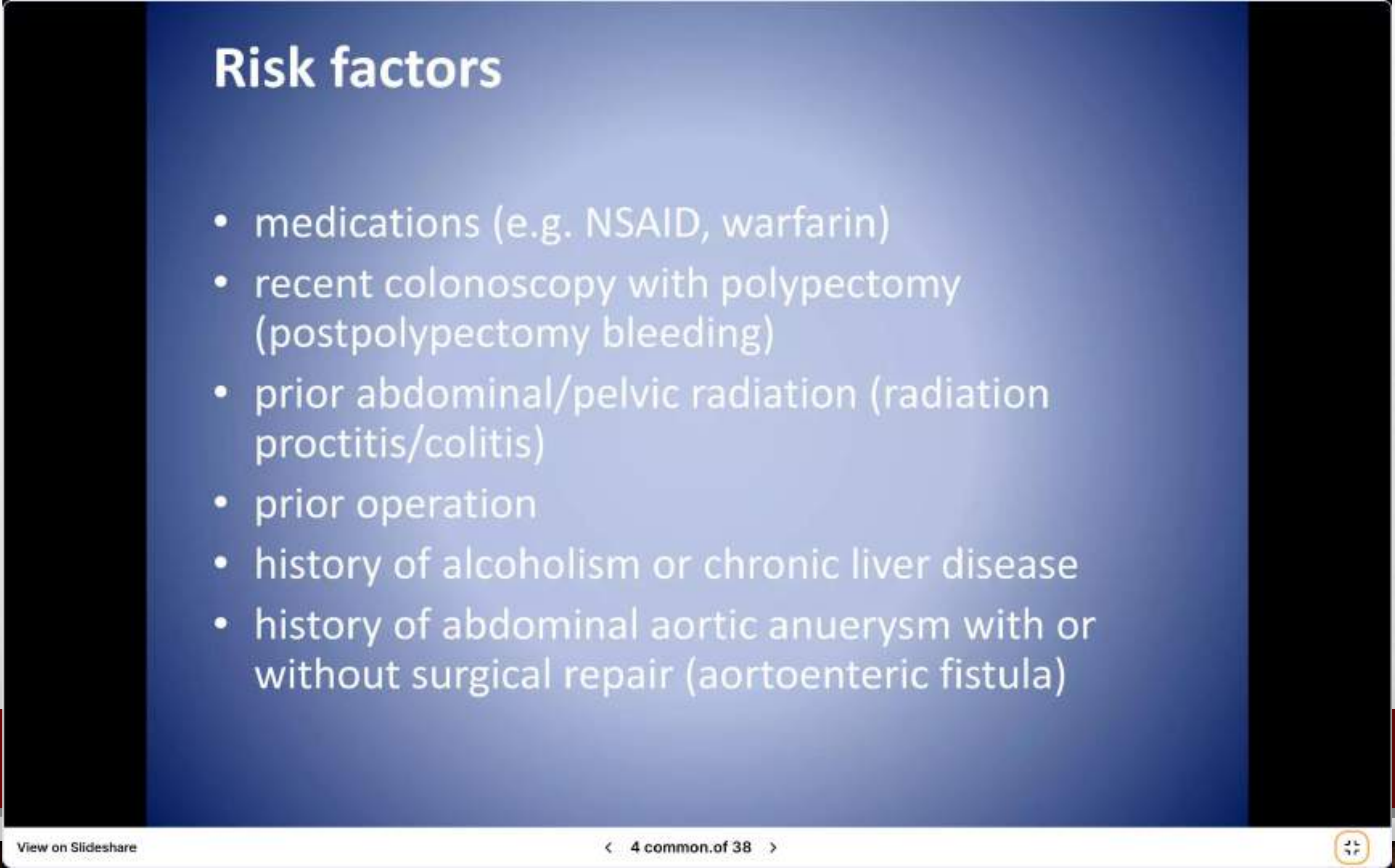
Presented by:

Students no. (31-50)

Round 1

Definition

- **Lower gastrointestinal bleeding (LGIB)** is defined as that occurring distal to the ligament of Treitz (i.e. from the jejunum, ileum, colon, rectum or anus) and presenting as either haematochezia (bright red blood/clots) or melaena


Risk factors
• medications (e.g. NSAID, warfarin)
• recent colonoscopy with polypectomy (postpolypectomy bleeding)
• prior abdominal/pelvic radiation (radiation proctitis/colitis)
• prior operation
• history of alcoholism or chronic liver disease
• history of abdominal aortic anuerysm with or without surgical repair (aortoenteric fistula)
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Causes

- Diverticular disease
- enterocolitis
  - infective
  - Crohn's disease
  - Ulcerative colitis
  - Ischemic colitis
- vascular malformation
  - vascular ectasia
  - Angiodysplasia
  - arteriovenous malformation (AVM)
- polyp
- tumour
- Vasculitides
- Portal hypertensive enteropathy or colopathy
- Meckel diverticulum
- ulcer
- Aorto-enteric fistula
- Anal fissure
- Haemorrhoids
- Perianal fistula

Differences between Crohn's disease (CD) and Ulcerative Colitis (UC)

	Ulcerative Colitis	Crohn's Disease
Affected area	Mainly affects the rectum and spread upwards in a continuous fashion	Mostly affects the small intestine especially the lower small intestine (ileum), but can occur anywhere in the GI tract
Developmental pattern	Spreads progressively from the rectum	Spreads unevenly - inflamed segments between normal healthy segments (skip lesions)
Ulcers	Ulceration only at first layer intestine	Ulceration at all layers of the intestine
Symptoms	Diarrhoea with blood and mucous	Abdominal pain, blood in stool is rare
Colon Canser risk	Higher risk compared to CD. Risk increase with the duration of disease and extent of disease.	

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Ischaemic colitis

- inflammation of the colon secondary to vascular insufficiency and ischaemia.
- It sometimes considered under the same spectrum of intestinal ischeamia.
- The severity and consequences of the disease are highly variable.


Age :
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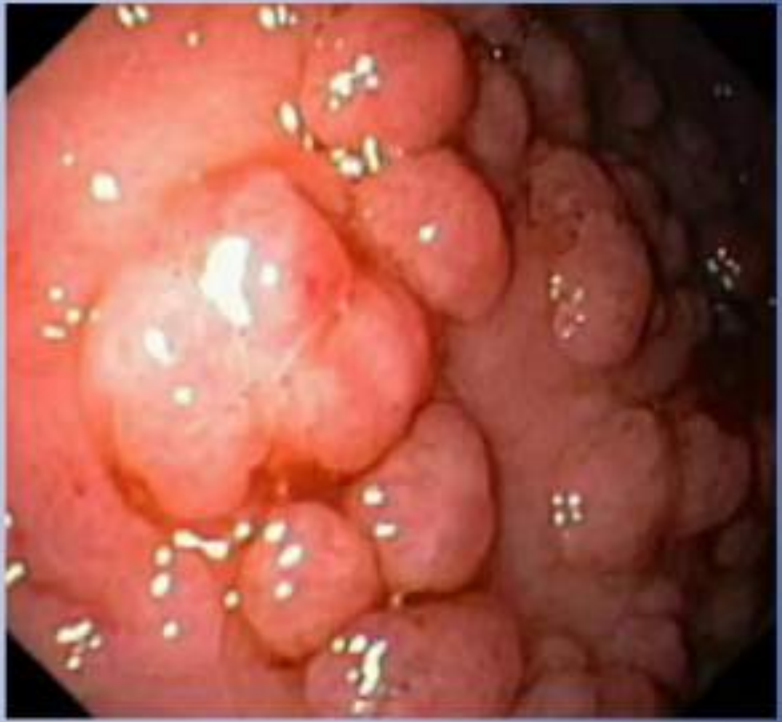
polyps

- Inflammatory
- Hamartomatous
- Neoplastic
- Hyperplastic

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